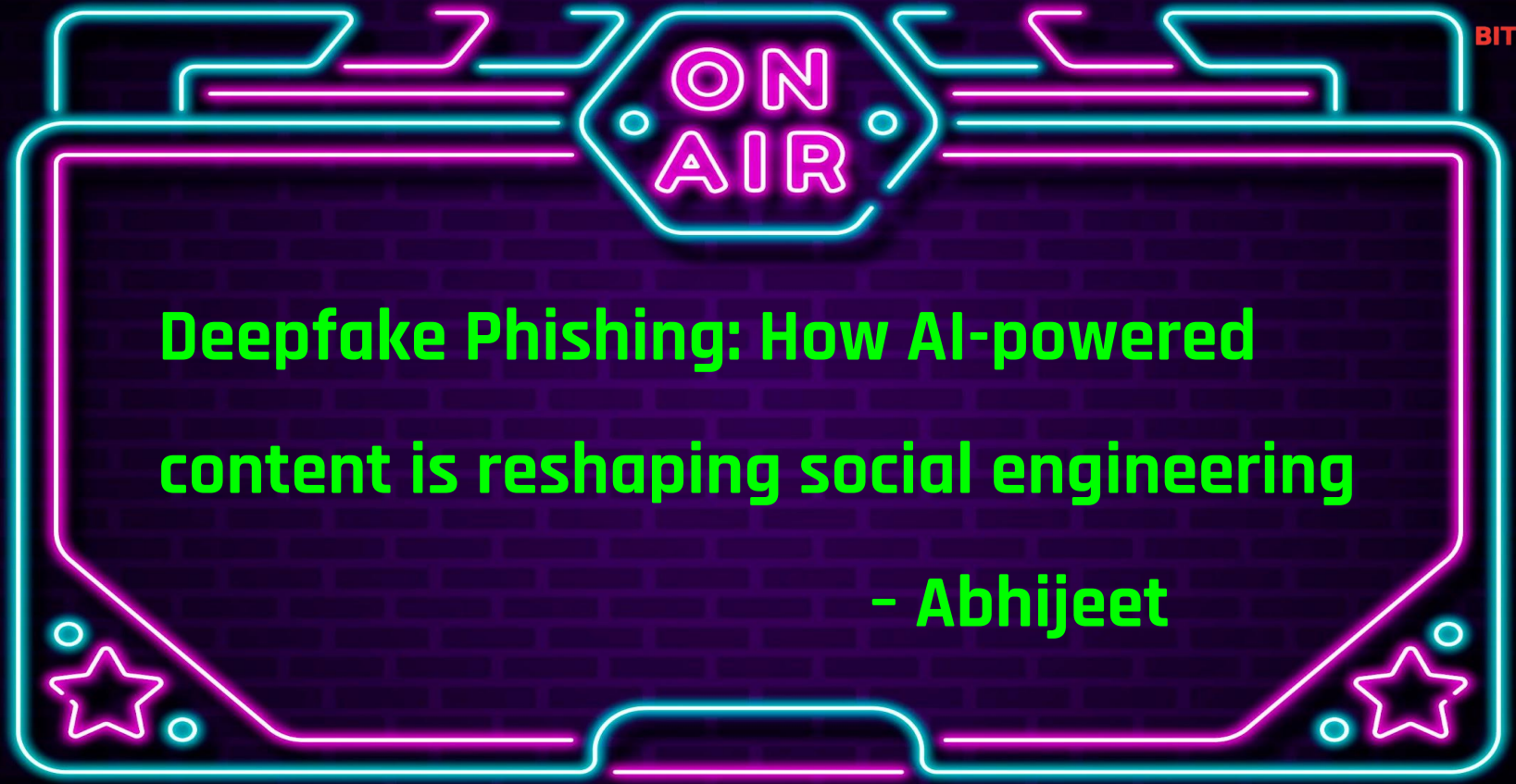


BIT-SEC-CON

Deepfake Phishing: How AI-powered content is reshaping social engineering

– Abhijeet

DISCLAIMER

- The techniques shown in this talk are offensive in nature
- It’s illegal to use the pictures/videos of anyone without their prior consent.
- Neither me nor the event organizers will be liable for any misuse of the talk materials.

WHOAMI

- Security Researcher & Tinkerer
  - Research areas include Adversarial TTPs, Linux systems, cloud infrastructure, and supply chain
- Full-time human, part-time homecook
- Current gig: Breaking things @CyberWarFare Labs

Social Engineering Lifecycle

1. Investigate

- Identify the victim.
- Gather background information.
- Select attack method.

2. Hook

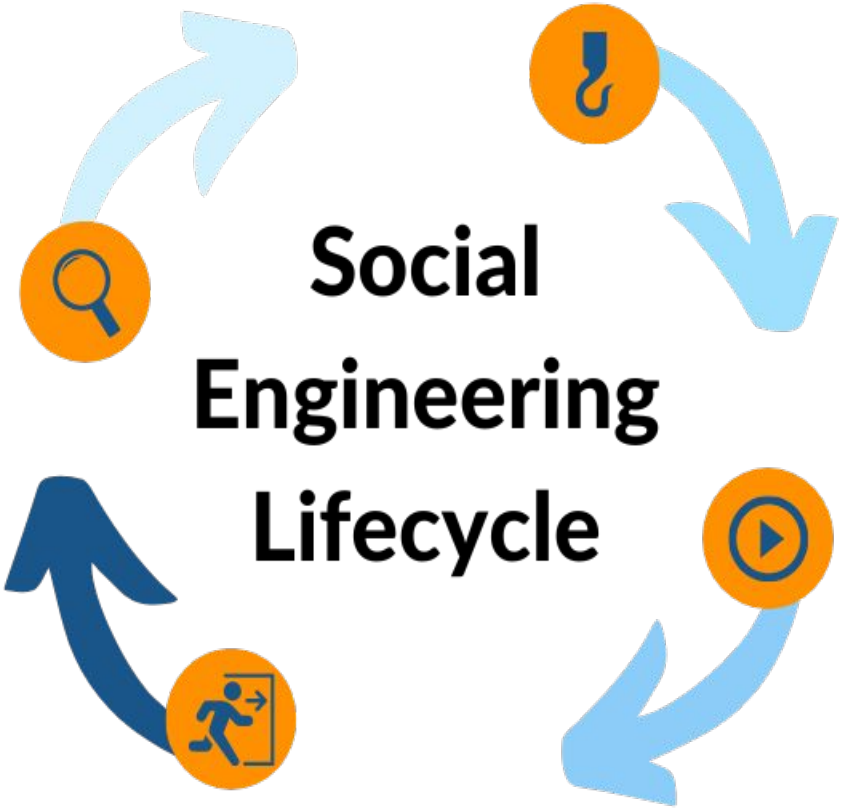
- Engaging the target.
- Spinning a story.
- Taking control of the interaction.

3. Play

- Expanding the foothold.
- Executing the attack.
- Disrupting business and/or siphoning data.

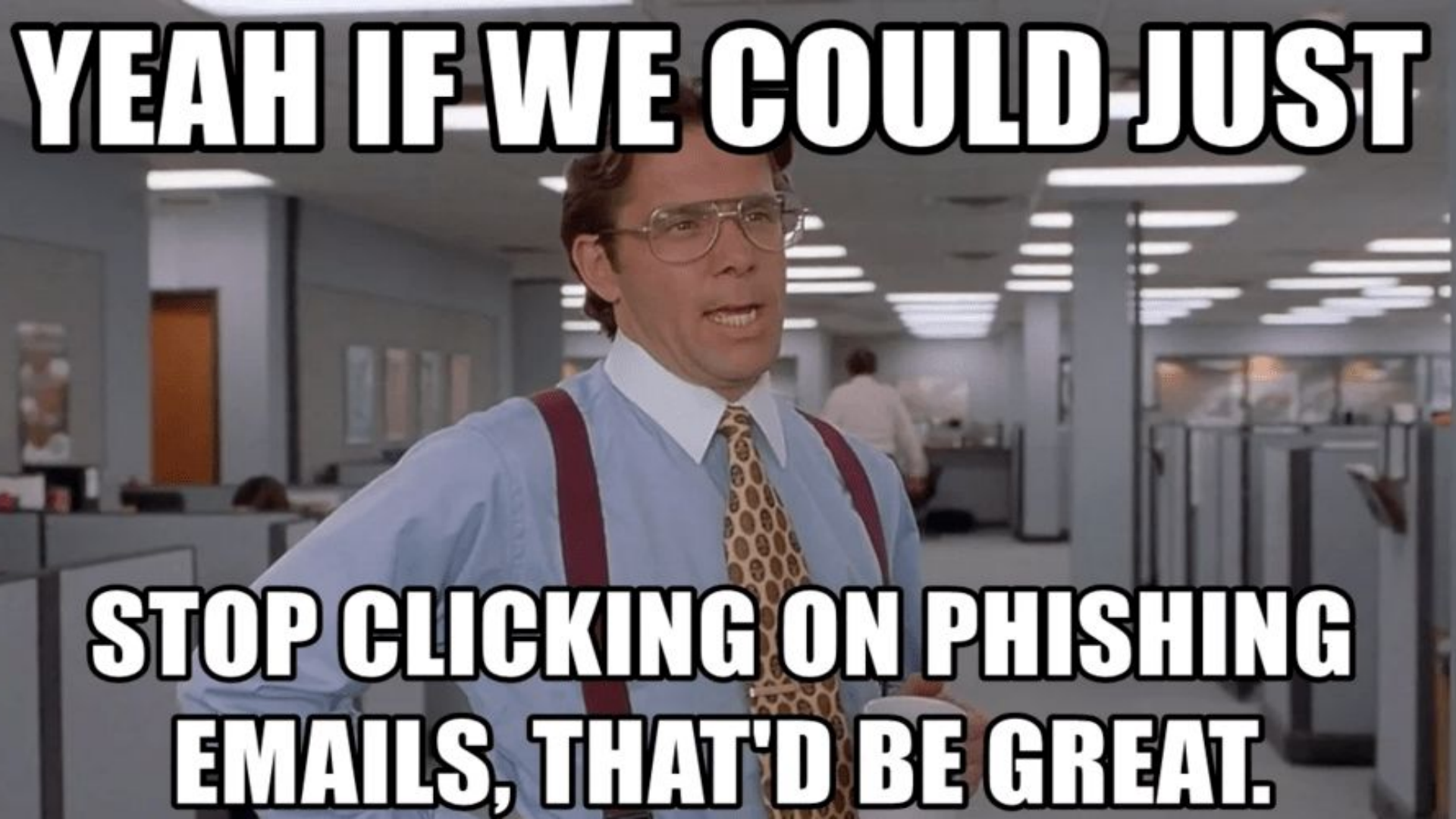
4. Exit

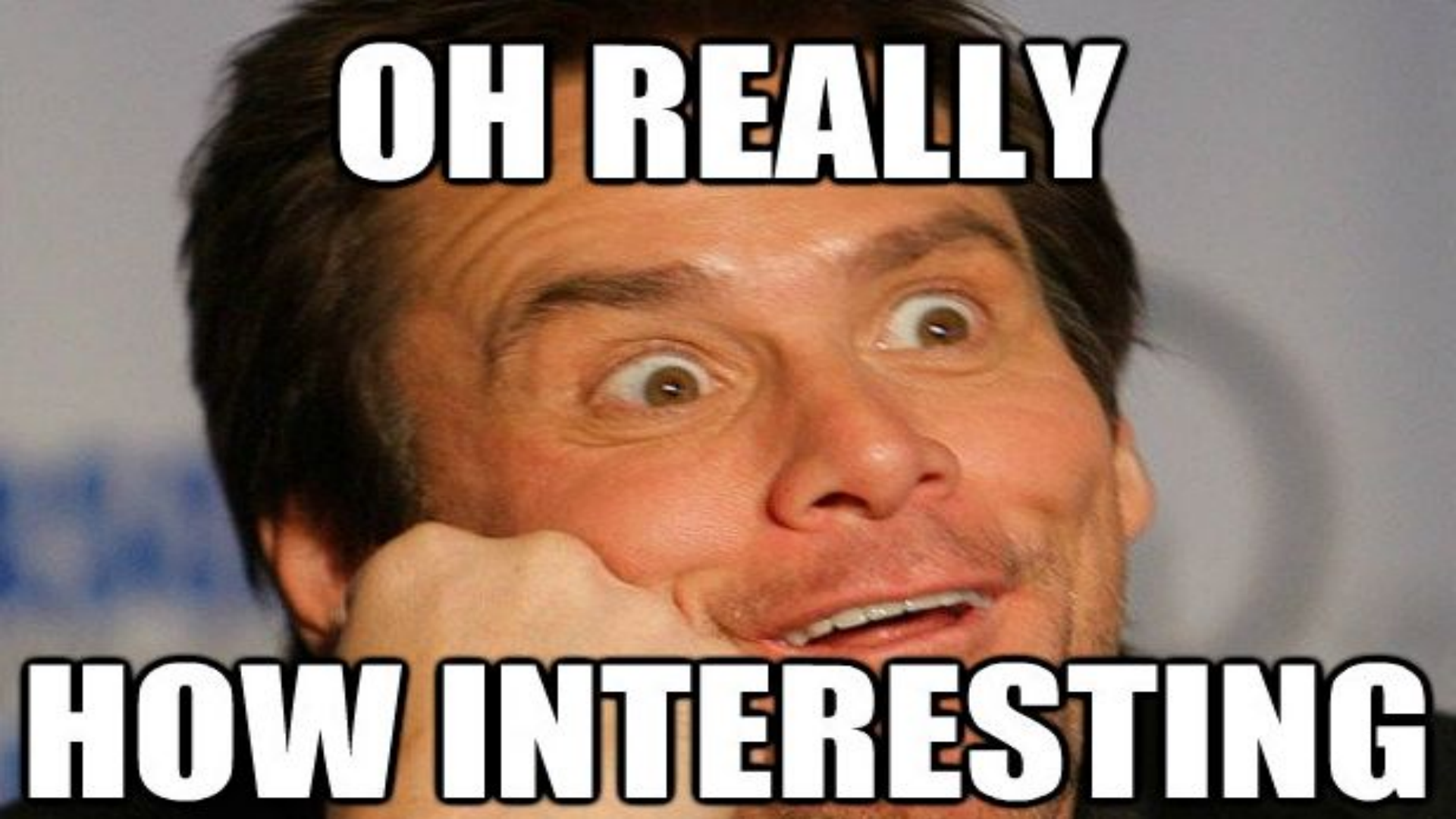
- Removing all traces of malware.
- Covering tracks.
- Bringing the play to a natural end.



PHISHING TYPES

- Traditional (Mass emails, fake sites, or malicious attachments)
- Spear (Targeted, personalized attacks)
- Smishing/Vishing (Phishing via SMS or phone calls)


YEAH IF WE COULD JUST
STOP CLICKING ON PHISHING EMAILS, THAT'D BE GREAT.


OH REALLY
HOW INTERESTING

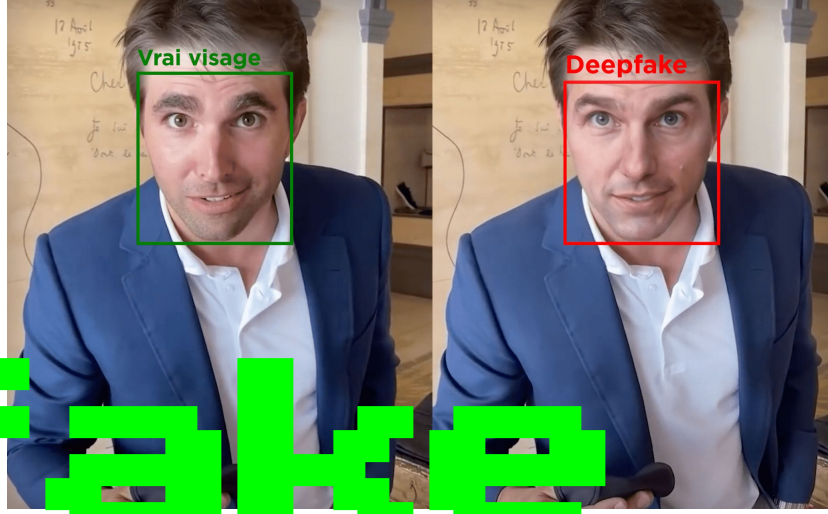

Here they come...


DEEP
FAKE

Now what the BLEEP is a deepfake?

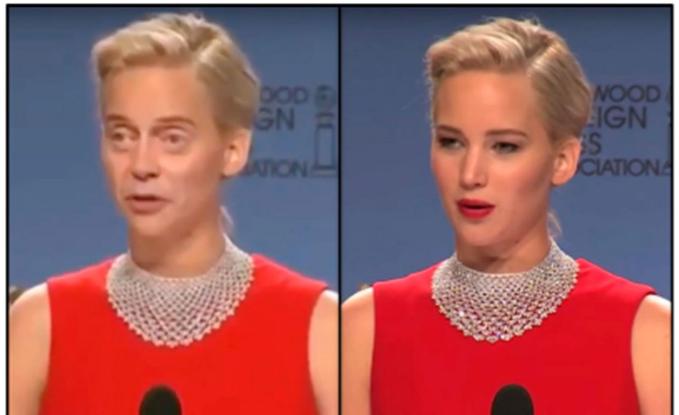


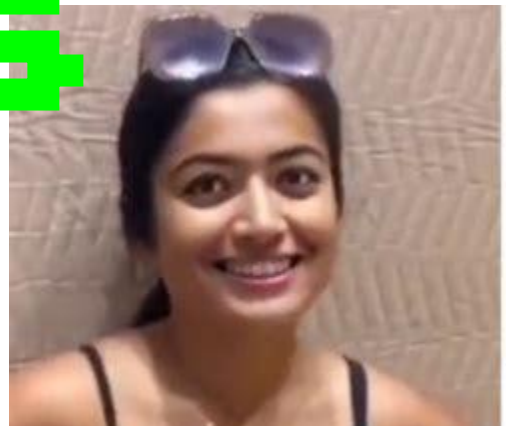




Deep fake

s








LET'S
WATCH VIDEOS
TOGETHER

Where Do They Come From?


You get a LLM

INTRODUCTION TO LLMS

- LLMs (Large Language Models) are subclass of artificial neural networks trained on massive datasets.
- Advanced LLMs comes under Generative Pre-trained Transformers (GPT).



INTRODUCTION TO LLMS

- Generative AI → text, images, voice, video.
- Types of models :-
  - Text-based (ChatGPT, Llama, Claude)
  - Voice cloning (ElevenLabs, VALL-E)
  - Text-to-video (Pika Labs, Runway, Sora)

DEEPFAKES 101

- Comes from combination of Deep (Deep Learning) and Fake.
- Uses AI, specifically ML algorithms to generate realistic audio/video/images of humans.



DEEPFAKES 101

- Deep Learning + Generative Models
- GANs (Generative Adversarial Networks)
- Diffusion Models (Stable Diffusion, Sora)
- Voice cloning → waveform + spectrogram models
- Video deepfakes → face swap + lip-sync + avatar gen


You must be new here

DEEPFAKES HISTORY

Computer Generated Imagery (CGI)

Researchers started using CGI to generate human images.

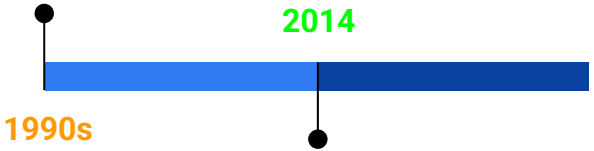


1990s

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Generative Adversarial Networks (GANs)

Ian Goodfellow developed the concept where two neural networks compete in a profit-loss game.

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Deepfake

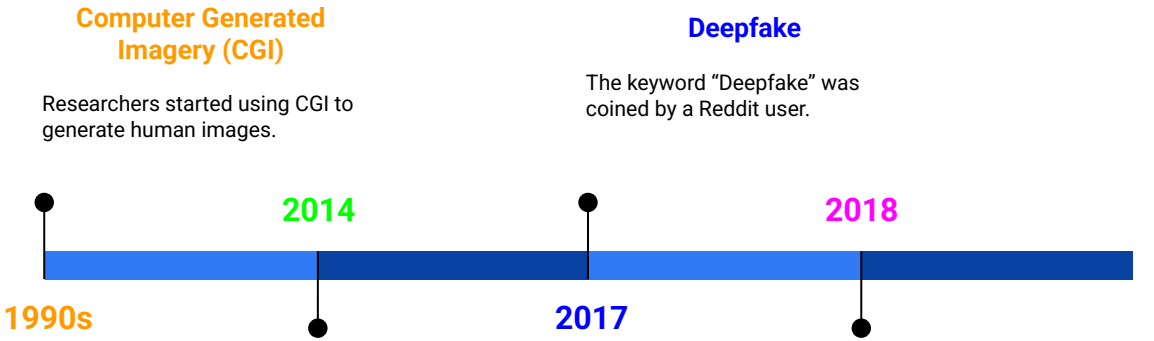
The keyword “Deepfake” was coined by a Reddit user.



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DEEPFAKES HISTORY



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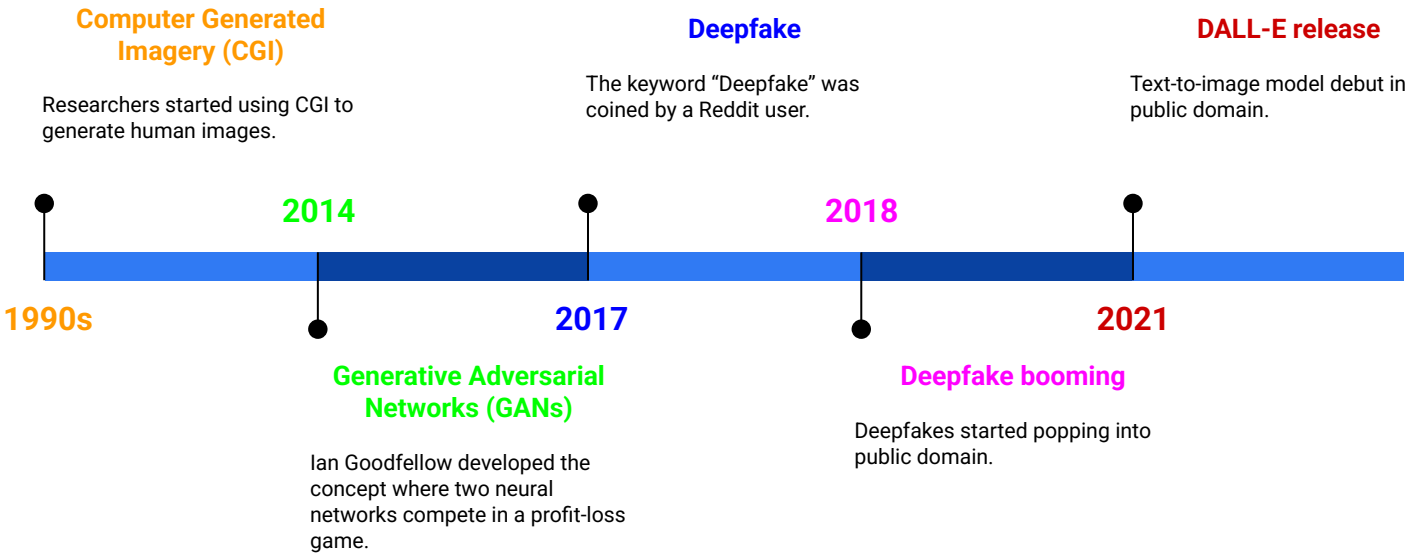
2017 — Deepfake

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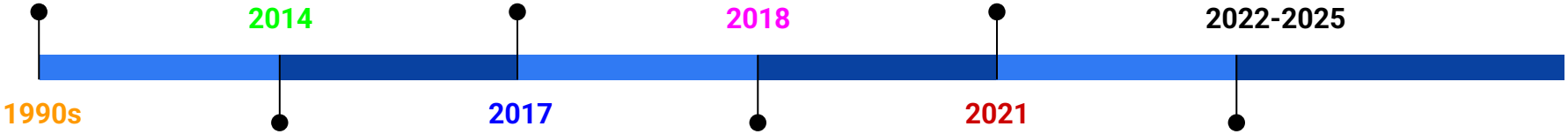
2018 — Deepfake booming

Deepfakes started popping into public domain.

DEEPFAKES HISTORY



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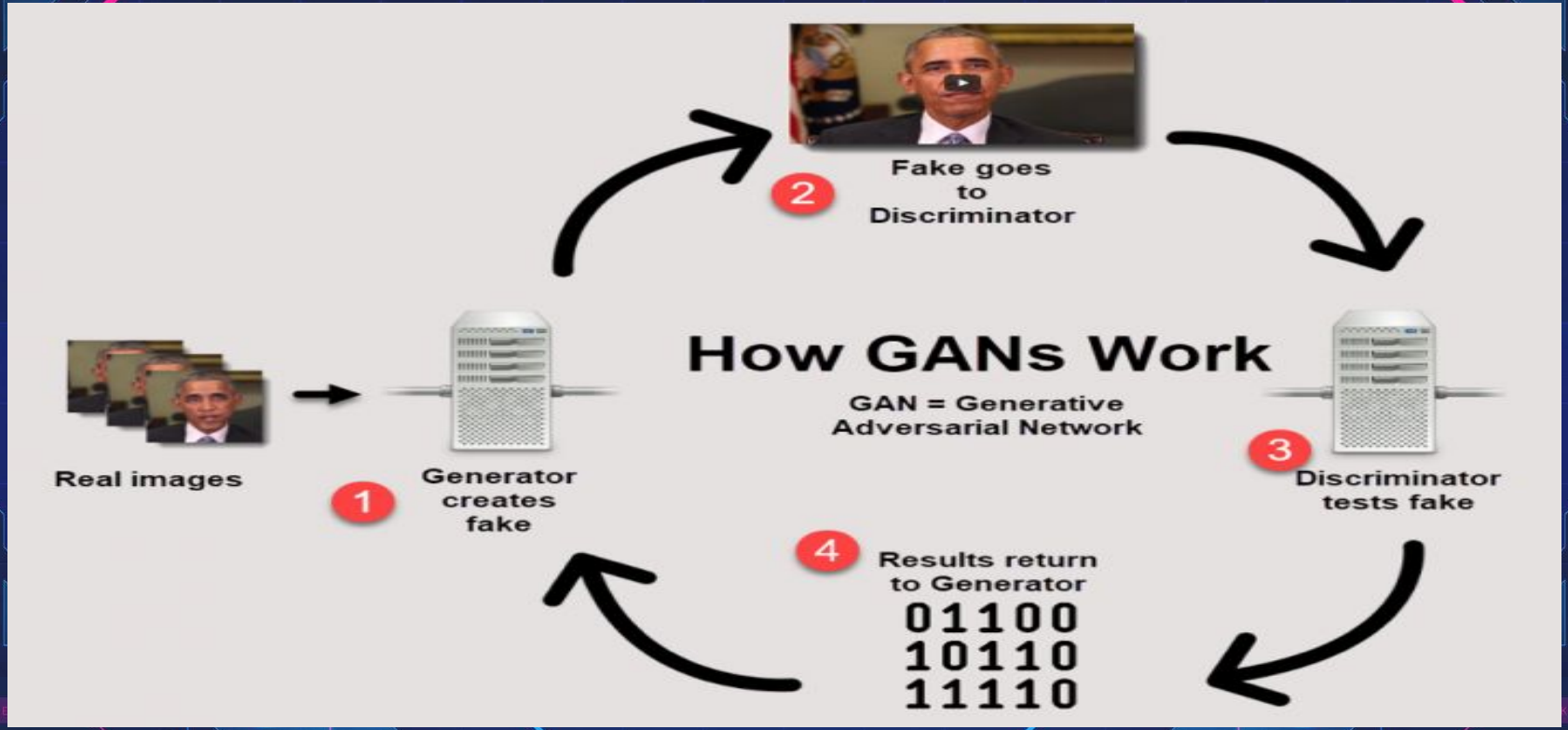
2021 — DALL-E release

Text-to-image model debut in public domain.

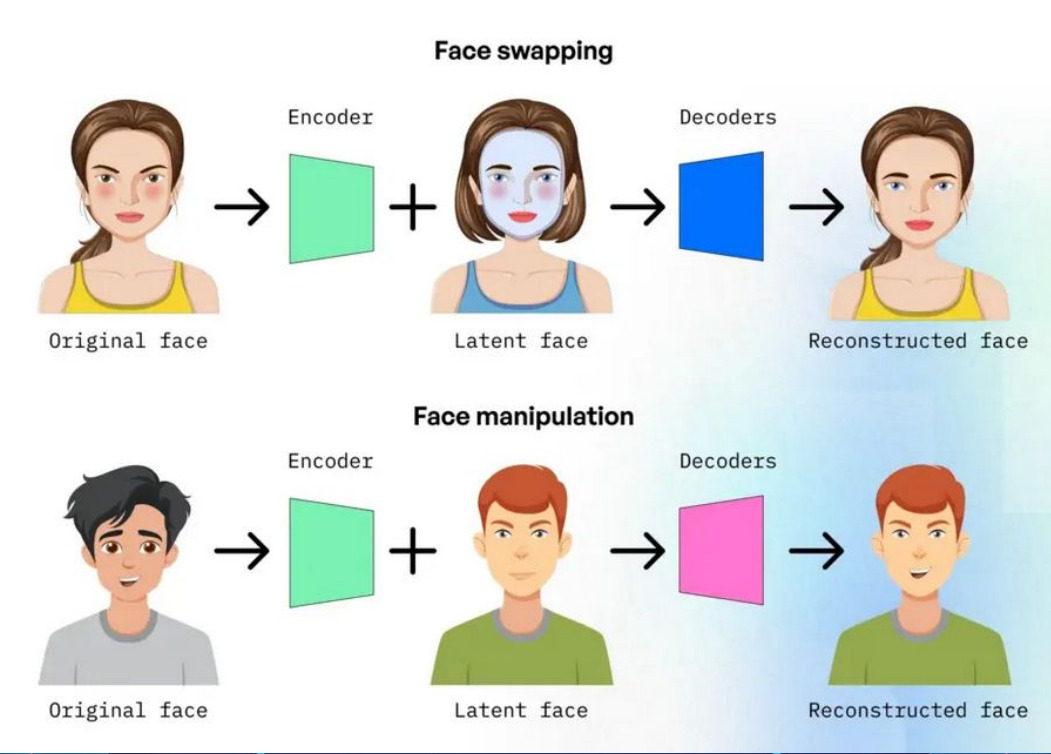
2022-2025 — Chaos

ChatGPT & other GenAI + LLM tools released, easing the creation of deepfakes for both technical & non-technical folks.

GENERATIVE ADVERSARIAL NETWORKS



DEEPFAKES TYPES



Ever used Instagram or Snapchat face filters?

WHY AI MAKES SOCIAL ENGINEERING BETTER?

- Generates personalized, fluent content instantly
- Removes language/grammar red flags
- Voice & face cloning enables synthetic trust
- Scales targeted phishing campaigns

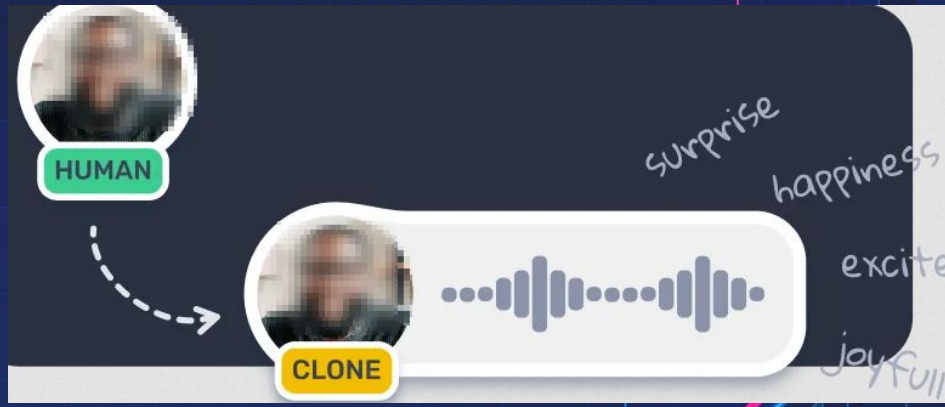
CASE STUDY 1: DPRK FAKE IT WORKER

- Threat Actor used stolen ID.
- AI-enhanced picture.
- Cleared video conferencing rounds.
- Got caught by SOC while loading malware.



CASE STUDY 2: VOICE CLONING CFO SCAM

- TA obtained voice samples.
- Created cloned voice.
- Hopped on a Zoom call with background noise & CEO's voice.
- Tricked finance team into transferring money.

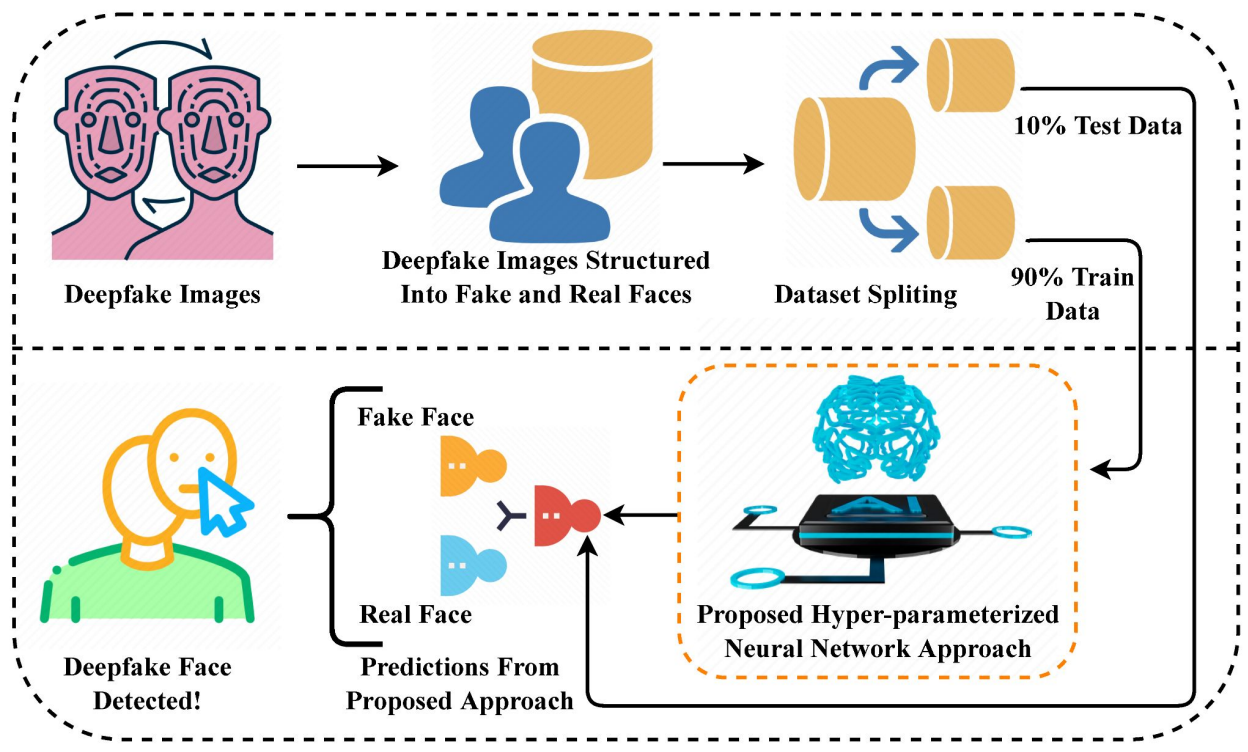


GENERATE YOUR OWN DEEPFAKES (GYOD)

- **Voice Cloning:-**
  - ElevenLabs
  - VALL-E
  - Respeecher
- **Video Deepfakes:-**
  - DeepFaceLab
  - DeepFaceLive
  - FaceSwap
  - Synthesia


LET'S DO THIS!

DEEPFAKE DETECTIONS



TAKEAWAYS

- Phishing has moved from text → voice → video
- AI removes traditional red flags
- Real-world deep fake phishing attacks have already caused millions in damages
- Defense requires multi-factor verification + awareness



Thank You



